Case Study 3: Student Performance Trend Analysis
Complete Python Source Code

```
subjects = ["Maths", "English", "Python"]
marks = [
    [55, 60, 72],
    [70, 68, 65],
    [35, 38, 39]
]
weak_limit = 40
final_total = 0
weak_subjects = []
print("STUDENT PERFORMANCE TREND")
for i in range(len(subjects)):
    first_mark = marks[i][0]
    final_mark = marks[i][2]
    change = final_mark - first_mark
    if change > 0:
        trend = "Improved"
    elif change < 0:
        trend = "Declined"
    else:
        trend = "Unchanged"
    if final_mark < weak_limit:
        weak_subjects.append(subjects[i])
    final_total += final_mark
    print(f"{subjects[i]}: first={first_mark}, final={final_mark}, change={change:+}, {trend}")
average = final_total / len(subjects)
print(f"Average final mark: {average:.2f}")
print("Weak subjects:", weak_subjects if weak_subjects else "None")
```